AI465 Research Project — Phase 1

MCPToolBench++: Evaluating Real MCP Tool Use

Area: Tool use and MCP

Primary paper: Fan et al., 2025 (arXiv:2508.07575)

U23AI052 | Individual work

Problem

01 MCP is becoming the standard tool interface for LLM agents.

02 Large-scale evaluation of real MCP tool use is still missing.

03 Existing tool-use benchmarks over-represent programming and math functions.

04 Real MCP servers vary in success rate, response format, and reliability.

05 One user query can require up to 10 dependent tool calls.

Method in Three Steps

01

Collect schemas

4,000+ MCP servers, 40+ categories

02

Generate QA

1,509 pairs across 6 domains

03

Evaluate

AST, AST DAG, Pass@K

Domains: browser, file system, search, map, finance, pay.

Chains up to 10 tools; multilingual queries.

AST checks function and parameters; Pass@K checks real execution.

What the Paper Reports

- Models tested: OpenAI, Claude, Qwen, Kimi, and others.
- Metrics: AST accuracy, AST DAG accuracy, Pass@K.
- Key finding: **high AST does not imply high Pass@K.**
- Causes: parameter errors, API errors, domain-specific issues.
- No cost, no latency, no variance, no failure taxonomy reported.

Gap 1 — Cost and Latency

- Schema tokens alone: 0.1K–0.6K per tool; 25K total across 6 domains.
- Without cost figures, accuracy gains cannot be judged against budget.

Reported	Not reported
AST, AST DAG, Pass@K	Tokens per task
Accuracy per model	Model calls per task
—	Wall-clock latency per task

Gap 2 — Variance

- One run per configuration, no seed reported.
- No standard deviation or confidence interval.
- Paper admits MCP server success rates vary.

Gap 3 — Failure Modes

- Failures mentioned anecdotally, not categorised.
- No split by tool selection vs. parameter vs. server error.
- Without this, the benchmark says **THAT** models fail, not **WHY**.

Objective

To evaluate the effect of tool-count context on MCP tool selection accuracy and failure composition for single-step and multi-step MCP tool-calling tasks on a 200-task stratified subset of MCPToolBench++ (100 single-step, 100 multi-step, across search, map, and finance domains), compared against the same model at a matched tool-count budget (5 vs. 20 vs. 50 tools in context), measured by AST accuracy, AST DAG accuracy, tokens per task, model calls per task, and wall-clock latency per task.

Traceable to Gap 1 and Gap 2; failure composition addresses Gap 3.

Feasible: 200 tasks, public benchmark, no training required.

Model reachable via free tier or local open-weight model.

Plan for Phase 2

01 Week 1: download and filter the 200-task subset.

02 Week 2: implement tool-count manipulation by truncating tool list.

03 Week 3: run each condition three times for variance.

04 Week 4: compute accuracy, token, call, latency, and failure metrics.

05 **Deliverable: short report with a failure taxonomy.**